

5

Colour

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5

COLOUR

COLOUR AND SCIENCE 211

5

COLOUR

COLOUR AND SCIENCE 212

Colour and Science

Colour is an essential element of multimedia,
 used in vector graphics, bitmapped images,
 video, animation and text.

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COLOUR AND SCIENCE 213

Colour science attempts to relate the subjective sensation of colour to measurable and reproducible physical phenomena.

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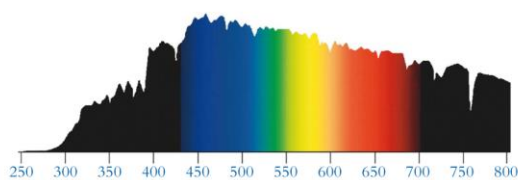
COLOUR AND SCIENCE 214

A spectral power distribution (SPD) is a description of how the intensity of light varies with its wavelength. An SPD is a good model of a colour, but is too cumbersome to work with in computer graphics.

5

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COLOUR AND SCIENCE 215



The spectral power distribution of daylight

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COLOUR AND SCIENCE 216

The tristimulus theory of colour implies that any colour can be produced by mixing suitable amounts of three additive primary colours.

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RGB COLOUR 217

RGB Colour

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COLOUR

RGB COLOUR 218

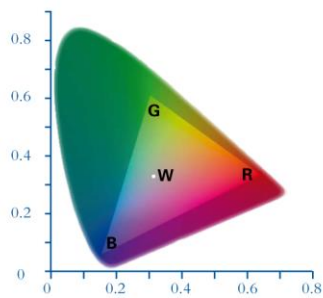
In RGB colour, the three primaries are standard shades of red, green and blue.

Only colours in the RGB gamut can be represented in this way.

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RGB COLOUR 219



The RGB colour gamut

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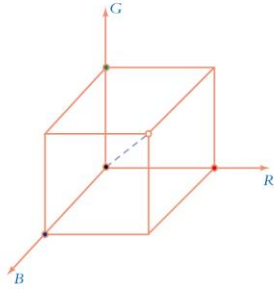
RGB COLOUR 220

Any colour is specified as three values (R , G , B) giving the relative proportions of the three primaries. This is often written as a 6-digit hexadecimal number, with R , G and B each being between 0 and 255, so a colour value occupies 24 bits.

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RGB COLOUR 221



The RGB colour space

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RGB COLOUR 222

The number of bits used to store a colour value – the colour depth – determines how many different colours can be represented. The use of lower colour depths leads to posterization and loss of image detail, but reduces file size.

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RGB COLOUR 223



A photograph in 24, 8 (top), 4 and 1 (bottom) bit colour

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RGB COLOUR 224



Detail of photograph in 24-bit (left) and 8-bit (right) colour

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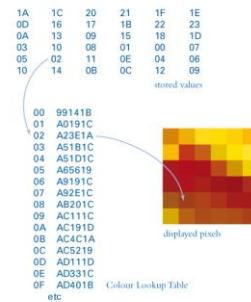
RGB COLOUR 225

In indexed colour, instead of storing a 24-bit colour value for each pixel, we use an 8-bit value which serves as an index into a colour table. The colour table contains the palette of colours used in the image.

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RGB COLOUR 226



Using a colour table

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RGB COLOUR 227



Images and their palettes

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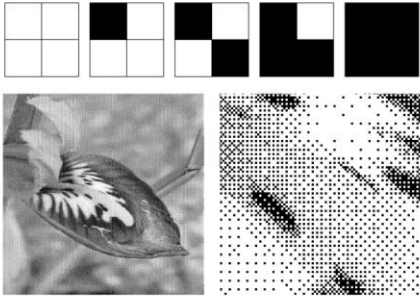
RGB COLOUR 228

Some colours from the original image may be missing from the palette. Dithering can be used to reduce the resulting posterization.

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RGB COLOUR 229

*Dithering in black and white*

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RGB COLOUR 230

*Dithering and posterization*

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OTHER COLOUR MODELS 231

Other Colour Models

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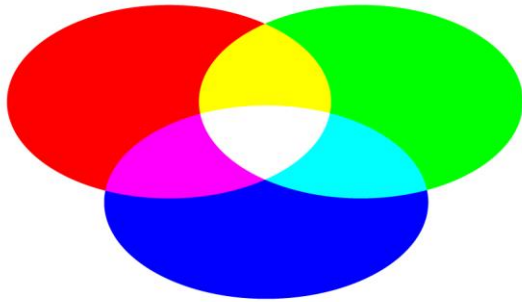
OTHER COLOUR MODELS 232

Cyan, magenta and yellow are the subtractive primaries. They are the complementary colours of red, green and blue, respectively.

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OTHER COLOUR MODELS 233



Complementary colours

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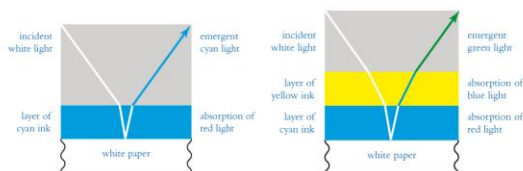
OTHER COLOUR MODELS 234

Thin layers of ink absorb some components of the incident light, so overlaying ink, as in printing processes, mixes colours subtractively.

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OTHER COLOUR MODELS 235



Coloured inks

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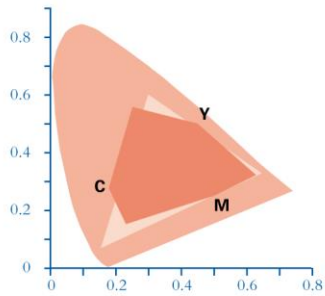
OTHER COLOUR MODELS 236

The CMYK colour gamut, corresponding to easily printable colours, is smaller than the RGB gamut, but some CMYK colours lie outside the RGB gamut.

5

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OTHER COLOUR MODELS 237



The CMYK gamut

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OTHER COLOUR MODELS 238

A colour can be identified by its hue, saturation and lightness.

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OTHER COLOUR MODELS 239

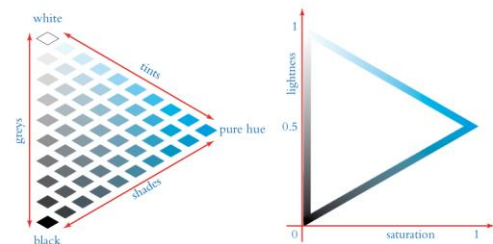
Tones of a single hue can be arranged two-dimensionally, with lightness increasing upwards, and saturation increasing from left to right.

Hues can be arranged around the rim of a colour wheel, with complementary colours opposite each other. A hue's value is the angle between its position on the wheel and the position of red.

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OTHER COLOUR MODELS 240



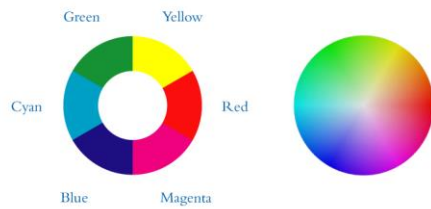
Tints, shades and tones

Saturation and lightness

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OTHER COLOUR MODELS 241

*A colour wheel*

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OTHER COLOUR MODELS 242

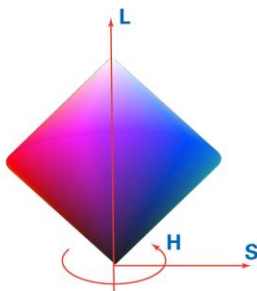
Hue, saturation and lightness can be combined into a three-dimensional double-cone. Any colour can be specified by its H, S and L components.

Filling the colour wheel

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OTHER COLOUR MODELS 243

*The HSL colour solid*

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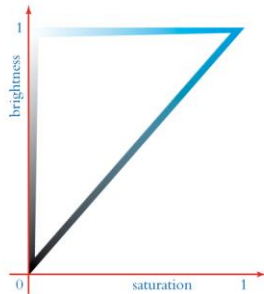
OTHER COLOUR MODELS 244

HSB is a variant of HSL, where the tones are arranged differently.

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OTHER COLOUR MODELS 245

*Saturation and brightness*

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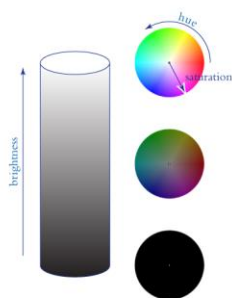
OTHER COLOUR MODELS 246

Both HSL and HSB are normally distorted into a cylindrical shape, so that they can be presented as colour pickers.

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OTHER COLOUR MODELS 247

*The HSB cylinder*

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COLOUR

OTHER COLOUR MODELS 248

*Colour pickers*

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OTHER COLOUR MODELS 249

Swatch libraries provide an alternative way of choosing colours.

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OTHER COLOUR MODELS 250

Colour spaces consisting of a brightness component and two colour differences are used in video. They allow chrominance sub-sampling to be used.

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OTHER COLOUR MODELS 251

The CIE $L^*a^*b^*$ and $L^*u^*v^*$ colour spaces are perceptually uniform and serve as device-independent reference models.

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CHANNELS AND COLOUR CORRECTION 252

Channels and Colour Correction

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CHANNELS AND COLOUR CORRECTION 253

The R, G and B components of each pixel can be stored as separate values.

The three arrays of values can be treated as greyscale images, called channels.

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CHANNELS AND COLOUR CORRECTION 254



An RGB colour image and its red, green and blue channels

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CHANNELS AND COLOUR CORRECTION 255

Making adjustments to the channels alters the colours of the image.

The colour balance, hue and saturation and colour replacement adjustments change the colour of the image as a whole.

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CHANNELS AND COLOUR CORRECTION 256



Hue and saturation adjustments

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CHANNELS AND COLOUR CORRECTION 257

*Colour replacement*

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CHANNELS AND COLOUR CORRECTION 258

Alpha channels can be treated as additional colour channels.

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CHANNELS AND COLOUR CORRECTION 259

Images using other colour models can also be separated into channels, which can be processed independently.

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CONSISTENT COLOUR 260

Consistent Colour

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CONSISTENT COLOUR 261

The colour properties of a monitor can be roughly summarized by the R, G and B chromaticities, white point and gamma.

Gamma approximately models the relationship between RGB values and light intensity.

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CONSISTENT COLOUR 262

If an image is stored in one device's colour space and displayed on a device with a different colour space, colours will not be reproduced accurately.

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CONSISTENT COLOUR 263



Inconsistent colours

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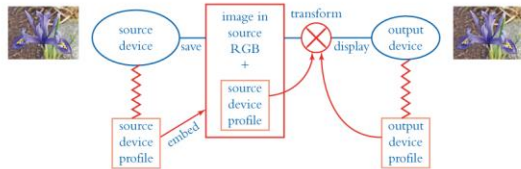
CONSISTENT COLOUR 264

If a colour profile that models the input device is embedded in the image file, it can be combined with a profile that models the output device to translate between the colour spaces and reproduce colours accurately.

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CONSISTENT COLOUR 265



Colour management with embedded profiles

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CONSISTENT COLOUR 266

Colours that are out of gamut should be reproduced consistently.

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CONSISTENT COLOUR 267

ICC colour profiles provide elaborate descriptions of the colour characteristics of a wide range of devices; they are used as a standard for colour management.

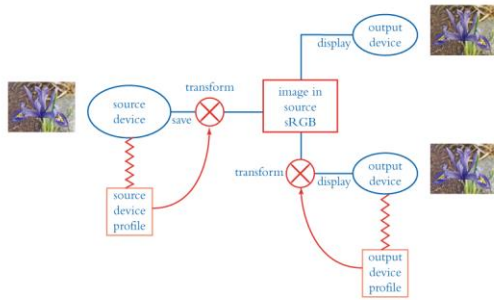
The success of colour management depends on having accurate profiles.

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CONSISTENT COLOUR 268

sRGB is intended as a standard device-independent colour space for monitors. It is used on the World Wide Web.



Use of the sRGB colour space